Why Ordinary Differentiation Fails on a Manifold

Deriving the Christoffel Symbols from Coordinate Transformations

Abstract

We derive the transformation law of the Christoffel symbols by demanding that the covariant derivative of a vector field transforms as a tensor. Starting from the observation that the ordinary partial derivative of a vector field is not a tensor, we show exactly how the Christoffel symbols must transform to cancel the unwanted terms. The derivation is self-contained and proceeds step-by-step from the coordinate transformation of a vector field to the final transformation law of the connection coefficients.

Introduction

In Euclidean space, the ordinary partial derivative $\partial_i Y^j$ of a vector field works perfectly. The coordinates are global, the basis vectors are constant, and the derivative has a clear geometric meaning.

But on a curved manifold, this simplicity vanishes. Coordinates are local, basis vectors change from point to point, and the partial derivative of a vector field depends on the coordinate system used. It is *not* a tensor.

This failure is not a mathematical inconvenience—it is a deep geometric fact. The partial derivative measures how components change, but it cannot distinguish between:

- A genuine change in the vector field itself.
- An apparent change caused by the twisting of the coordinate grid.

To fix this, we must introduce a new object: the *covariant derivative.* We define:

$$\nabla_i Y^j = \partial_i Y^j + \Gamma^j_{ik} Y^k.$$

The extra term Γ^j_{ik} is designed to compensate for the coordinate dependence of the partial derivative. But what is this Γ? How must it transform so that $\nabla_i Y^j$ becomes a true tensor?

In this note, we answer exactly that question. We begin with the transformation law of a vector field, compute how the partial derivative changes under a coordinate transformation, and then determine the transformation law of Γ^j_{ik} by demanding that the covariant derivative transforms as a tensor.

The result will be:

$$\boxed{\Gamma^{j'}_{i'k'} = \frac{\partial x^i}{\partial x^{i'}} \frac{\partial x^j}{\partial x^{j'}} \frac{\partial x^k}{\partial x^{k'}} \Gamma^j_{ik} - \frac{\partial x^i}{\partial x^{i'}} \frac{\partial x^k}{\partial x^{k'}} \frac{\partial^2 x^{j'}}{\partial x^i \partial x^k}}$$

The Christoffel symbols are **not** tensors. They transform inhomogeneously, with an extra term involving second derivatives of the coordinate transformation. This inhomogeneous term is precisely what cancels the coordinate-dependent part of $\partial_i Y^j$, making the covariant derivative a true geometric object.

1 The Question We Want to Answer

Let M be a smooth manifold. We have a vector field

$$Y \in \mathfrak{X}(M).$$

In local coordinates $x^1, \dots, x^n$,

$$Y = Y^j \frac{\partial}{\partial x^j},$$

where:

- Y^j are the components of the vector field.
- $\frac{\partial}{\partial x^j}$ are the coordinate basis vectors.

We want to calculate the derivative of Y. The natural idea is:

$$\partial_i Y^j = \frac{\partial Y^j}{\partial x^i}.$$

In ordinary Euclidean space this is enough. But on a manifold we must ask:

Does this derivative represent a geometric object, independent of the chosen coordinates?

For that, we must check its transformation behaviour.

2 Change of Coordinates

Consider another coordinate system:

$$x^1, \dots, x^n \quad \text{and} \quad x^{1'}, \dots, x^{n'}.$$

The point on the manifold has not changed. Only the description changes:

$$x^i \to x^{i'}.$$

The vector field Y is the same geometric object:

$$Y = Y^j \frac{\partial}{\partial x^j}$$

and also:

$$Y = Y^{j'} \frac{\partial}{\partial x^{j'}}.$$

3 Transformation of the Vector Components

The basis vectors transform according to the chain rule:

$$\frac{\partial}{\partial x^{j'}} = \frac{\partial x^j}{\partial x^{j'}} \frac{\partial}{\partial x^j}.$$

Therefore:

$$Y = Y^{j'} \frac{\partial x^j}{\partial x^{j'}} \frac{\partial}{\partial x^j}.$$

Comparing with

$$Y = Y^j \frac{\partial}{\partial x^j},$$

we get:

$$Y^j = Y^{j'} \frac{\partial x^j}{\partial x^{j'}}.$$

Therefore:

$$\boxed{Y^{j'} = \frac{\partial x^{j'}}{\partial x^j} Y^j}$$

This is the normal vector transformation law.

4 Differentiate the Vector Field

We now calculate in the new coordinates:

$$\partial_{i'} Y^{j'}.$$

Insert the transformation law:

$$\partial_{i'} Y^{j'} = \partial_{i'} \left(\frac{\partial x^{j'}}{\partial x^j} Y^j \right).$$

Now the important step: the derivative acts on both factors. Using the product rule:

$$\partial_{i'} Y^{j'} = \left(\partial_{i'} \frac{\partial x^{j'}}{\partial x^j} \right) Y^j + \frac{\partial x^{j'}}{\partial x^j} \partial_{i'} Y^j.$$

This equation contains the whole problem.

5 Interpret the Two Terms

The second term is what we would like:

$$\frac{\partial x^{j'}}{\partial x^j} \partial_{i'} Y^j.$$

Using

$$\partial_{i'} = \frac{\partial x^i}{\partial x^{i'}} \partial_i,$$

we get:

$$\frac{\partial x^{j'}}{\partial x^j} \frac{\partial x^i}{\partial x^{i'}} \partial_i Y^j.$$

Rearranging:

$$\frac{\partial x^i}{\partial x^{i'}} \frac{\partial x^{j'}}{\partial x^j} \partial_i Y^j.$$

This has exactly the transformation law of a $(1, 1)$-tensor.
But the first term is:

$$\left(\partial_{i'} \frac{\partial x^{j'}}{\partial x^j} \right) Y^j.$$

This contains:

$$\frac{\partial x^i}{\partial x^{i'}}\frac{\partial^2 x^{j'}}{\partial x^i \partial x^j}.$$

It depends on how the coordinate system bends.
Therefore:

$$\boxed{\partial_i Y^j \text{ is not a tensor.}}$$

6 The Problem We Must Solve

We want an object:

$$\nabla_i Y^j$$

which transforms like:

$$\nabla_{i'} Y^{j'} = \frac{\partial x^i}{\partial x^{i'}}\frac{\partial x^{j'}}{\partial x^j}\nabla_i Y^j.$$

That means the unwanted term must be cancelled.
So we modify the ordinary derivative:

$$\boxed{\nabla_i Y^j = \partial_i Y^j + \Gamma^j_{ik} Y^k}$$

The new term is designed to compensate for the coordinate change.

7 Write the Transformation Explicitly

In the new coordinates:

$$\nabla_{i'} Y^{j'} = \partial_{i'} Y^{j'} + \Gamma^{j'}_{i'k'} Y^{k'}.$$

Using our Step 5 result for $\partial_{i'} Y^{j'}$:

$$\nabla_{i'} Y^{j'} = \frac{\partial x^i}{\partial x^{i'}}\frac{\partial x^{j'}}{\partial x^j}\partial_i Y^j + \frac{\partial x^i}{\partial x^{i'}}\frac{\partial^2 x^{j'}}{\partial x^i \partial x^j} Y^j + \Gamma^{j'}_{i'k'} Y^{k'}.$$

8 Transform $Y^{k'}$

Since $Y^{k'} = \frac{\partial x^{k'}}{\partial x^k} Y^k$:

$$\nabla_{i'} Y^{j'} = \frac{\partial x^i}{\partial x^{i'}}\frac{\partial x^{j'}}{\partial x^j}\partial_i Y^j + \frac{\partial x^i}{\partial x^{i'}}\frac{\partial^2 x^{j'}}{\partial x^i \partial x^j} Y^j + \Gamma^{j'}_{i'k'}\frac{\partial x^{k'}}{\partial x^k} Y^k.$$

9 Set This Equal to the Tensor Transformation Law

We require:

$$\nabla_{i'} Y^{j'} = \frac{\partial x^i}{\partial x^{i'}}\frac{\partial x^{j'}}{\partial x^j}\left(\partial_i Y^j + \Gamma^j_{ik} Y^k\right).$$

Expanding:

$$\nabla_{i'} Y^{j'} = \frac{\partial x^i}{\partial x^{i'}}\frac{\partial x^{j'}}{\partial x^j}\partial_i Y^j + \frac{\partial x^i}{\partial x^{i'}}\frac{\partial x^{j'}}{\partial x^j}\Gamma^j_{ik} Y^k.$$

10 Equate the Two Expressions

From Step 8 and Step 9:

$$\frac{\partial x^i}{\partial x^{i'}}\frac{\partial x^{j'}}{\partial x^j}\partial_i Y^j + \frac{\partial x^i}{\partial x^{i'}}\frac{\partial^2 x^{j'}}{\partial x^i \partial x^j}Y^j + \Gamma^{j'}_{i'k'}\frac{\partial x^{k'}}{\partial x^k}Y^k = \frac{\partial x^i}{\partial x^{i'}}\frac{\partial x^{j'}}{\partial x^j}\partial_i Y^j + \frac{\partial x^i}{\partial x^{i'}}\frac{\partial x^{j'}}{\partial x^j}\Gamma^j_{ik}Y^k.$$

Cancel the common first term:

$$\frac{\partial x^i}{\partial x^{i'}}\frac{\partial^2 x^{j'}}{\partial x^i \partial x^j}Y^j + \Gamma^{j'}_{i'k'}\frac{\partial x^{k'}}{\partial x^k}Y^k = \frac{\partial x^i}{\partial x^{i'}}\frac{\partial x^{j'}}{\partial x^j}\Gamma^j_{ik}Y^k.$$

11 Rename Dummy Indices

In the first term on the left, rename $j \to k$:

$$\frac{\partial x^i}{\partial x^{i'}}\frac{\partial^2 x^{j'}}{\partial x^i \partial x^k}Y^k + \Gamma^{j'}_{i'k'}\frac{\partial x^{k'}}{\partial x^k}Y^k = \frac{\partial x^i}{\partial x^{i'}}\frac{\partial x^{j'}}{\partial x^j}\Gamma^j_{ik}Y^k.$$

Factor out Y^k:

$$\left(\frac{\partial x^i}{\partial x^{i'}}\frac{\partial^2 x^{j'}}{\partial x^i \partial x^k} + \Gamma^{j'}_{i'k'}\frac{\partial x^{k'}}{\partial x^k}\right)Y^k = \frac{\partial x^i}{\partial x^{i'}}\frac{\partial x^{j'}}{\partial x^j}\Gamma^j_{ik}Y^k.$$

Since this holds for arbitrary Y^k:

$$\frac{\partial x^i}{\partial x^{i'}}\frac{\partial^2 x^{j'}}{\partial x^i \partial x^k} + \Gamma^{j'}_{i'k'}\frac{\partial x^{k'}}{\partial x^k} = \frac{\partial x^i}{\partial x^{i'}}\frac{\partial x^{j'}}{\partial x^j}\Gamma^j_{ik}.$$

12 Solve for $\Gamma^{j'}_{i'k'}$

$$\Gamma^{j'}_{i'k'}\frac{\partial x^{k'}}{\partial x^k} = \frac{\partial x^i}{\partial x^{i'}}\frac{\partial x^{j'}}{\partial x^j}\Gamma^j_{ik} - \frac{\partial x^i}{\partial x^{i'}}\frac{\partial^2 x^{j'}}{\partial x^i \partial x^k}.$$

Multiply both sides by $\frac{\partial x^k}{\partial x^{k'}}$:

$$\boxed{\Gamma^{j'}_{i'k'} = \frac{\partial x^i}{\partial x^{i'}}\frac{\partial x^j}{\partial x^{j'}}\frac{\partial x^k}{\partial x^{k'}}\Gamma^j_{ik} - \frac{\partial x^i}{\partial x^{i'}}\frac{\partial x^k}{\partial x^{k'}}\frac{\partial^2 x^{j'}}{\partial x^i \partial x^k}}$$

13 The Meaning of This Formula

The Christoffel symbols transform with **two types of terms**:

Term	Origin
$\frac{\partial x^i}{\partial x^{i'}}\frac{\partial x^j}{\partial x^{j'}}\frac{\partial x^k}{\partial x^{k'}}\Gamma^j_{ik}$	Tensor-like part (homogeneous)
$-\frac{\partial x^i}{\partial x^{i'}}\frac{\partial x^k}{\partial x^{k'}}\frac{\partial^2 x^{j'}}{\partial x^i \partial x^k}$	Inhomogeneous part (cancels the extra term from $\partial_i Y^j$)

14 Summary

Object	Transformation Law
$\partial_i Y^j$	$\frac{\partial x^i}{\partial x^{i'}}\frac{\partial x^{j'}}{\partial x^j}\partial_i Y^j + \frac{\partial x^i}{\partial x^{i'}}\frac{\partial^2 x^{j'}}{\partial x^i \partial x^j} Y^j$
Γ^j_{ik}	$\frac{\partial x^i}{\partial x^{i'}}\frac{\partial x^j}{\partial x^{j'}}\frac{\partial x^k}{\partial x^{k'}}\Gamma^j_{ik} - \frac{\partial x^i}{\partial x^{i'}}\frac{\partial x^k}{\partial x^{k'}}\frac{\partial^2 x^{j'}}{\partial x^i \partial x^k}$
$\nabla_i Y^j = \partial_i Y^j + \Gamma^j_{ik} Y^k$	$\frac{\partial x^i}{\partial x^{i'}}\frac{\partial x^{j'}}{\partial x^j}\nabla_i Y^j$ (tensor)

15 The Key Insight

The Christoffel symbols are **not** tensors. Their inhomogeneous transformation law is precisely designed to cancel the inhomogeneous term in the transformation of $\partial_i Y^j$, making the covariant derivative $\nabla_i Y^j$ a true tensor.

The covariant derivative exists because the Christoffel symbols transform in exactly the right way to cancel the coordinate-dependent part of the partial derivative.

Conclusion

We have derived the transformation law of the Christoffel symbols from first principles. The key steps were:

1. Start with the coordinate transformation of a vector field: $Y^{j'} = \frac{\partial x^{j'}}{\partial x^j} Y^j$.
2. Differentiate to find that $\partial_i Y^j$ transforms with an extra term involving second derivatives of the coordinate transformation.
3. Define the covariant derivative $\nabla_i Y^j = \partial_i Y^j + \Gamma^j_{ik} Y^k$.
4. Demand that $\nabla_i Y^j$ transforms as a tensor.
5. Solve for the transformation law of Γ^j_{ik}.

The result shows that the Christoffel symbols are not geometric objects in their own right—they are *compensating fields* that depend on the coordinate system. Their job is to absorb the coordinate dependence of the partial derivative, leaving behind the true geometric object: the covariant derivative.

This is the fundamental reason why differential geometry on curved manifolds requires a connection. The ordinary derivative is not enough; we must introduce a rule for comparing vectors at different points. That rule is encoded in the Christoffel symbols, and their transformation law is forced upon us by the requirement of geometric objectivity.